OXFORD ECONOMICS

UK NEUTRAL INTEREST RATE – ONLINE APPENDIX

Dr Daniel Harenberg, Lead Global Economist

Edward Allenby, UK Economist

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Short model summary

We build on state-of-the-art academic research

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A Model of Secular Stagnation: Theory and Quantitative Evaluation[†]

By Gauti B. Eggertsson, Neil R. Mehrotra, and Jacob A. Robbins*

This paper formalizes and quantifies the secular stagnation hypothesis, defined as a persistently low or negative natural rate of interest leading to a chronically binding zero lower bound (ZLB). Output-inflation dynamics and policy prescriptions are fundamentally different from those in the standard New Keynesian framework. Using a 56-period quantitative life cycle model, a standard calibration to US data delivers a natural rate ranging from −1.5 percent to −2 percent, implying an elevated risk of ZLB episodes for the foreseeable future. We decompose the contribution of demographic and technological factors to the decline in interest rates since 1970 and quantify changes required to restore higher rates. (JEL E12, E23, E31, E32, E43, E52)

The zero lower bound (ZLB) on the short-term nominal interest rate became a binding constraint in the United States in 2008 in the midst of the financial crisis. Accordingly, low interest rates are often tied to that event, and it may seem natural to presume that as the crisis of 2008 moves into the rear view mirror so too will the low interest rate environment. This perspective seems vindicated by the recent interest rate increases by the Federal Reserve in the fourth quarter of 2016. However, as shown in Figure 1, the low interest rate in 2008 was not just an anomaly that arose solely because of the financial crisis. Instead, it is the culmination of a 25-year trend across major industrial economies. In Japan, rates have been zero since the mid-1990s and remain there today. Furthermore, while recent increases in the Federal Funds rate may give rise to optimism in the United States, rates remain so low should raise at zero in Europe as of this writing. The fact that rates remain so low should raise

* Eggertsson: Department of Economics, Brown University, 64 Waterman Street, Providence, RI 02906 (email: gauti_eggertsson@brown.edu); Mehrotra: Department of Economics, Brown University, 64 Waterman Street, Providence, RI 02906 (email: neil_mehrotra@brown.edu); Robbins: Department of Economics, Brown University, 64 Waterman Street, Providence, RI 02906 (email: jacob_robbins@brown.edu). This paper replaces an earlier version of a paper released in 2014 under the title "A Model of Secular Stagnation." We thank the Institute for New Economic Thinking for financial support. We also thank Olivier Blanchard, John Cochrane, Michael Dotsey, Emmanuel Farhi, Jesus Fernandez-Villaverde, Pablo Guerron-Quintana, Benjamin Keen, Pat Kehoe, John Leahy, Emi Nakamura, Paolo Pesenti, and Lawrence Summers for helpful discussions, and conference and seminar participants at the Bank of Canada, Bank of England, Bank for International Settlements, Bank of Japan, Bank of Finland, Boston University, Brown, Central Bank of Austria, Cornell, Chicago Booth, Georgetown, European Central Bank, Rutgers, the Federal Reserve Banks of Boston, Dallas, Minneapolis, and New York, Harvard, LSE, LUISS Guido Carli, NBER Summer Institute MEFM and EFG meetings, Northwestern, Princeton, Royal Economic Society meetings, and UTDT for comments. We would also like to thank Alex Mechanick for excellent research assistance.
† Go to https://doi.org/10.1257/mac.20170367 to visit the article page for additional materials and author disclosure statement(s) or to comment in the online discussion forum.

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https://doi.org/10.1257/mac.20170367

- **We extend Eggertsson, Mehrotra & Robbins (2019)**
- Very innovative and influential paper
- "New Keynesian Overlapping Generations Model"
- EMR consider 7 secular drivers. We add
  - International capital flows
  - Net migration
  - Stock of securities held by central bank
- We extend the horizon from 2016 to 2050
- For details, see our white paper (p. 7-9)

A New Keynesian Overlapping Generations Model

- Overlapping generations model
  - Captures economic impact of ageing populations
  - E.g., life expectancy rising faster than retirement age ➔ more savings
  - Possibility of persistently negative real interest rate
- Interacts with New Keynesian frictions
  - Effective lower bound ➔ neutral rate can fall below policy rates
  - Modern liquidity trap ➔ central bank cannot sufficiently stimulate demand with policy rates
  - Nominal rigidities ➔ when inflation slows, wage growth responds more sluggishly
  - Monopolistic competition ➔ rising market power leads to rising profits accruing to owners

Overview of the model (1/2)

- **Dynamic general equilibrium model (with incomplete markets and rational agents)**
- **Asset markets**: nominal government bonds, loans to households, equity in firms, cash
- **Households**: earn labour and pension income, make consumption and investment decisions under borrowing constraints, supply labour inelastically, leave bequests, are forward-looking.
- **Firms**: produce differentiated final goods, intermediate goods, investment goods. Employ labour and capital to maximise profits, given an exogenous process for total factor productivity. Monopolistic competition in final goods leads to profits that are distributed to households.
- **Government**: issues nominal bonds, taxes labour and capital, runs a pay-as-you-go social security system

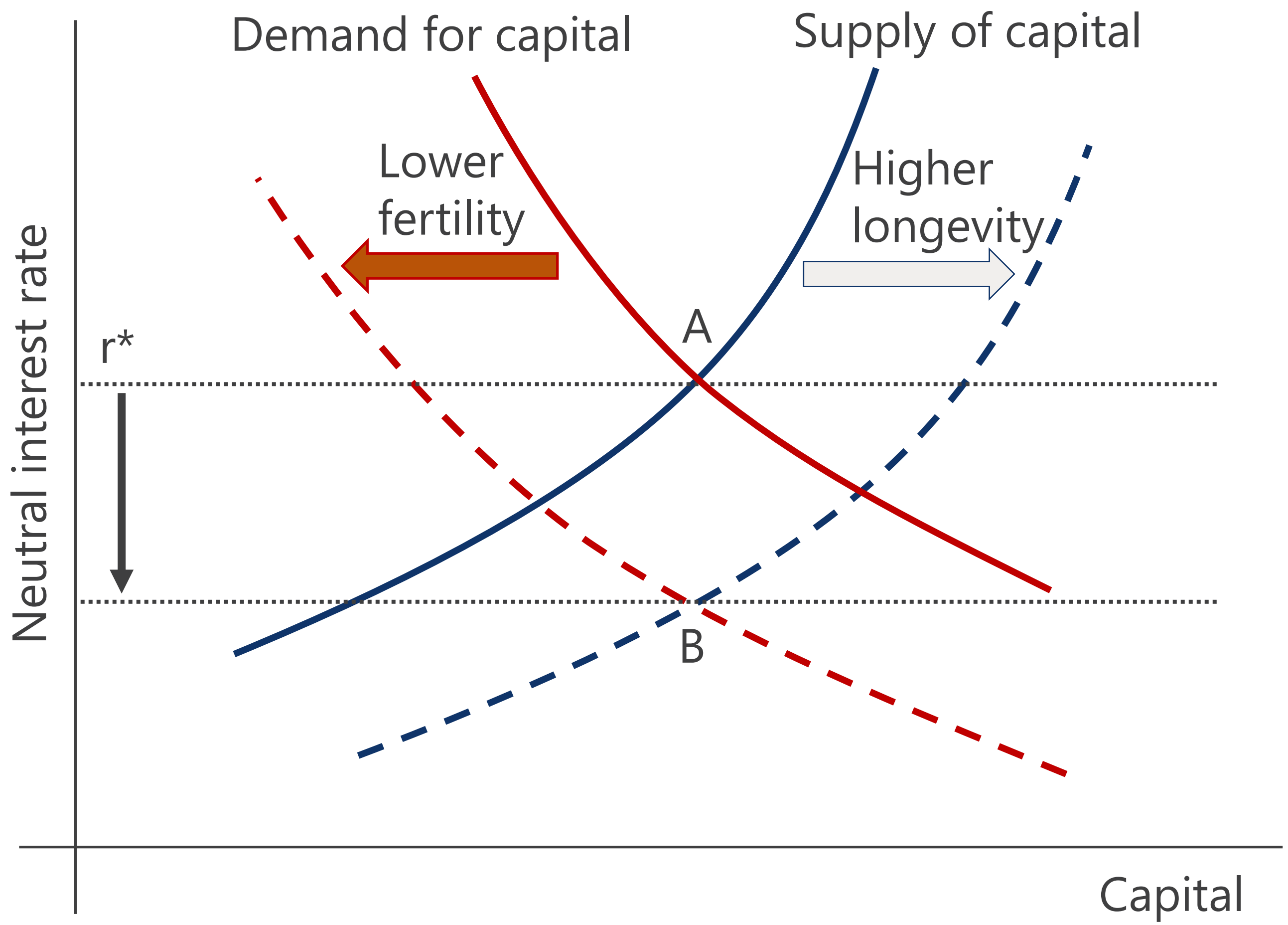
Overview of the model (2/2)

- **Central bank**: sets nominal policy with Taylor rule, constrained by zero lower bound (ZLB).
  - We introduce stock of securities held under QE programs: Central bank expands balance sheet by buying government bonds according to historic data and exogenous forecast
  - Affects bond neutral rate via market segmentation / preferred habitat assumption
- **Rest of world:** buys domestic assets, offers foreign assets, uses proceeds to finance imports
  - We use Oxford Economics' forecast of net foreign assets (NFA). See our related work on net foreign assets in an international OLG model (based on Auclert et al., 2022).
- **Nominal wage rigidities**: When inflation is at or above target, nominal wages rise at the rate of inflation. But when inflation is below target, nominal wage growth is sticky and does not decrease as quickly.

The real neutral interest rate, r*

- **Laubach-Williams**: r* is the rate consistent with closed output gap and inflation at target
  - Estimates of output gap often at odds with official sources. Can't analyse underlying forces.
- **Here**: r* is the rate that balances demand and supply in the capital market
  - Consistent with closed output gap and inflation at target
  - Allows us to intuitively and yet formally discuss underlying forces
  - Determined by joint behaviour of all agents and the secular forces
  - We focus on r* consistent with central bank Taylor rule (in addition, model separately computes an economy-wide natural rate)
- Nominal neutral interest rate = r* + inflation expectations

Determination of the real neutral interest rate



- Neutral rate equilibrates supply and demand in the capital market
- Aggregate capital supply and demand by adding behaviour of all agents
- Higher longevity ➔ more savings
- Lower fertility ➔ less loan demand
- Higher public debt ➔ more demand for capital
- Similar logic for other secular forces
- In model: dynamic over time

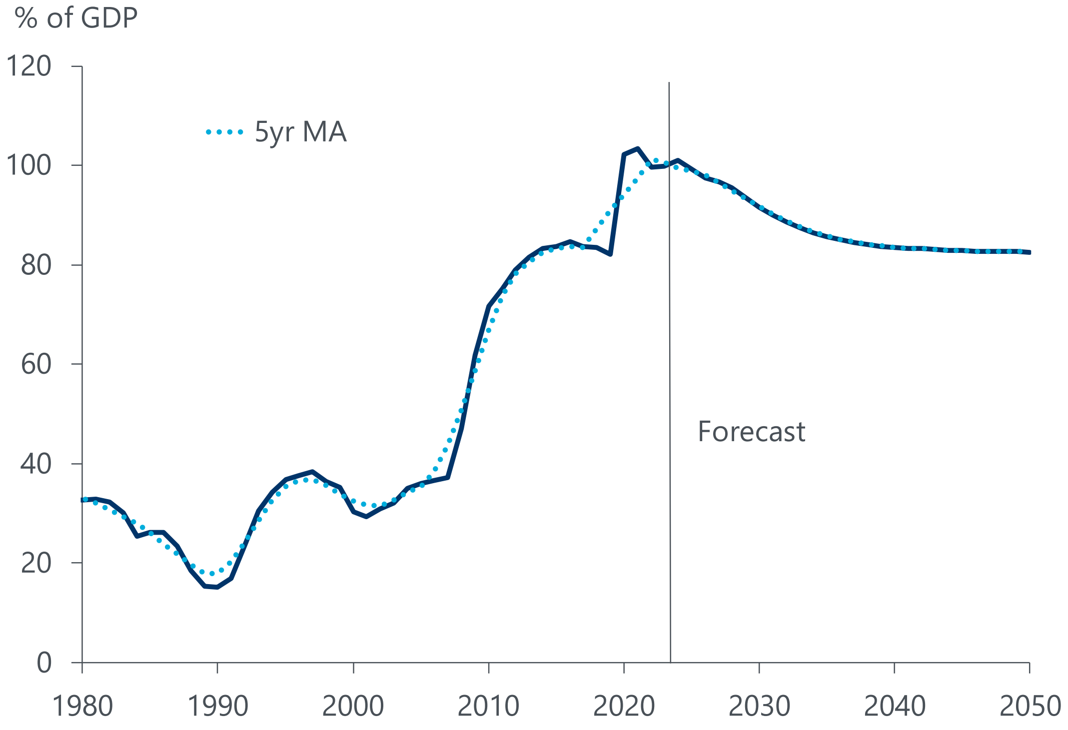
Directly related Oxford Economics research briefings

- Eurozone: Long-term forces cap the bloc's neutral interest rate, Jan 2024
- US: The neutral rate has risen, but by less than most think, Oct 2023
- US: Why the neutral rate will fall below pre-pandemic lows, Sep 2022
- Global: Structural forces to depress policy rates in the medium term, Jul 2022
- Global: Why an ageing population doesn't mean soaring inflation, May 2022
- Global: Why global demographics are not inflationary, Feb 2022
- Eurozone: Long-term forces to keep interest rates and inflation low, Nov 2021
- Europe: Rising wealth depresses interest rates in the UK & EZ, Sep 2021
- Global: Demographic forces will swell global financial imbalances, Jul 2021
- Global: Net foreign assets after the pandemic, Jul 2021
- US: A low neutral policy rate with a 1% handle, Jan 2021
- Eurozone: Secular stagnation is real. Getting out of it will be hard, Oct 2020
- Secular Stagnation in the Eurozone, Oct 2020
- Eurozone: ECB's one-size-fits-all monetary policy is a problem, Feb 2020
- Eurozone: Less than zero: Lagarde's interest rate challenge, Jul 2019

The secular forces – baseline assumptions

Secular forces: Government debt

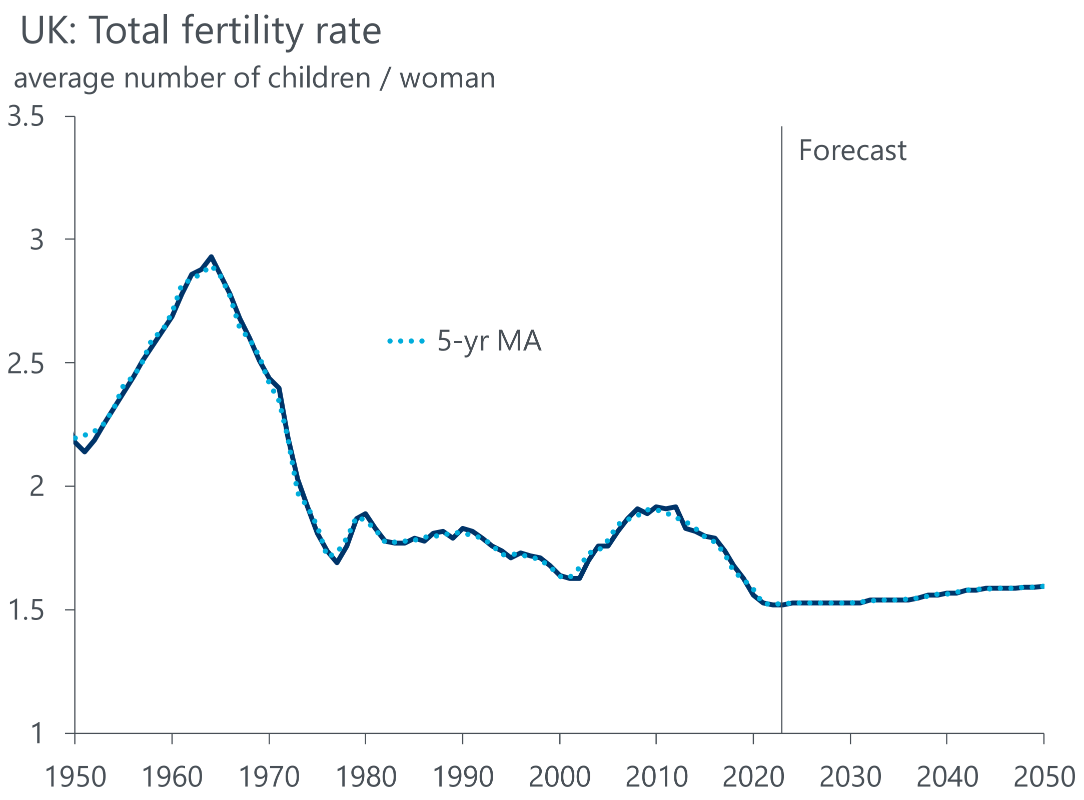
UK: Government debt (excl. pension liabilities)



Source: Oxford Economics, Haver Analytics, Bank of England

- Outstanding government debt securities
- Based on OE forecast of total government debt, available in our Global Economic Databank
- But excluding pension liabilities, which are modeled separately, because much is an unfunded (pay-as-you-go) system, which does not directly affect supply of outstanding government securities.
- Includes institutional holdings (to account for QE and NFA)
- Impact on r*: upward in history, downward in forecast

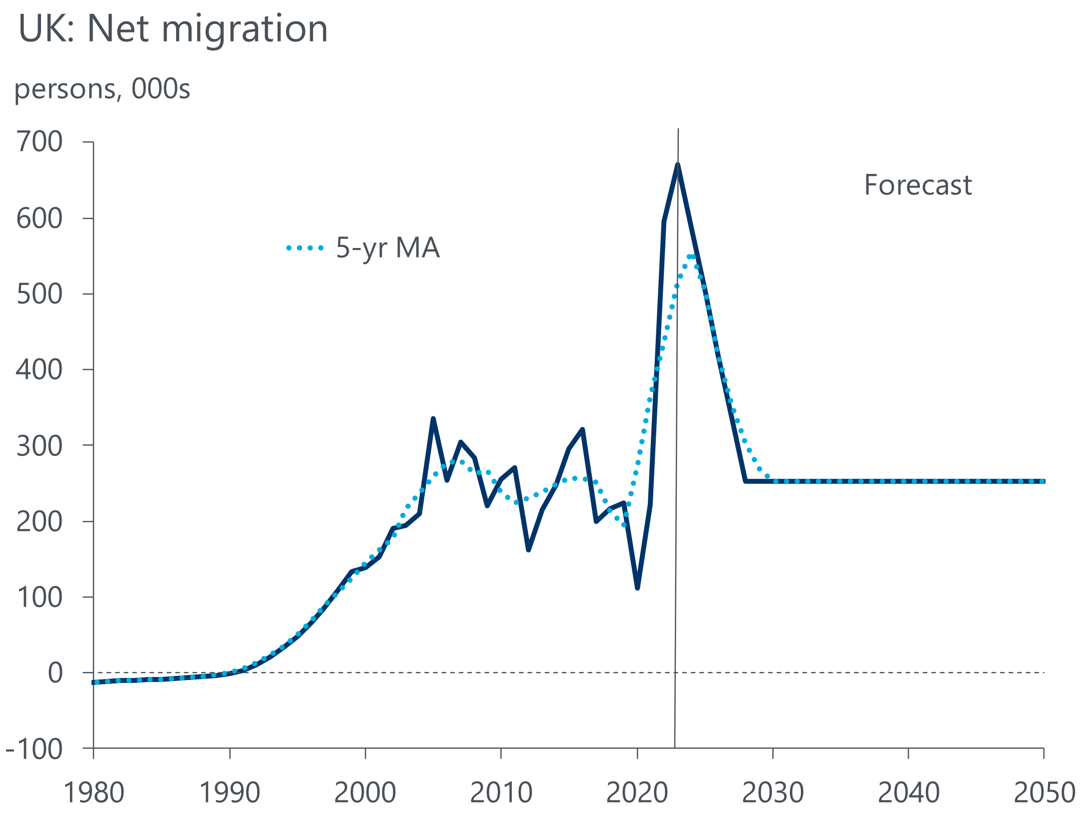
Secular forces: Total fertility rate



Source: Oxford Economics/Haver Analytics/ONS

- Average number of children / woman, based on age-specific fertility rates in each year
- Forecast based on ONS baseline projection
- TFR history back to 1950, so that simulated population in 1980 close to true population
- Strong negative impact on r* in multiple ways:
  - Fewer young who ask for loans in capital market, lowering capital demand
  - Smaller labor force, capital becomes relatively abundant, stock and bond returns fall
  - Weaker private consumption growth, firms invest less, further reduction in loan demand.
  - Lower potential output growth.
- Impact on r* with 20-year delay, when agents enter labour force

Secular forces: Net migration

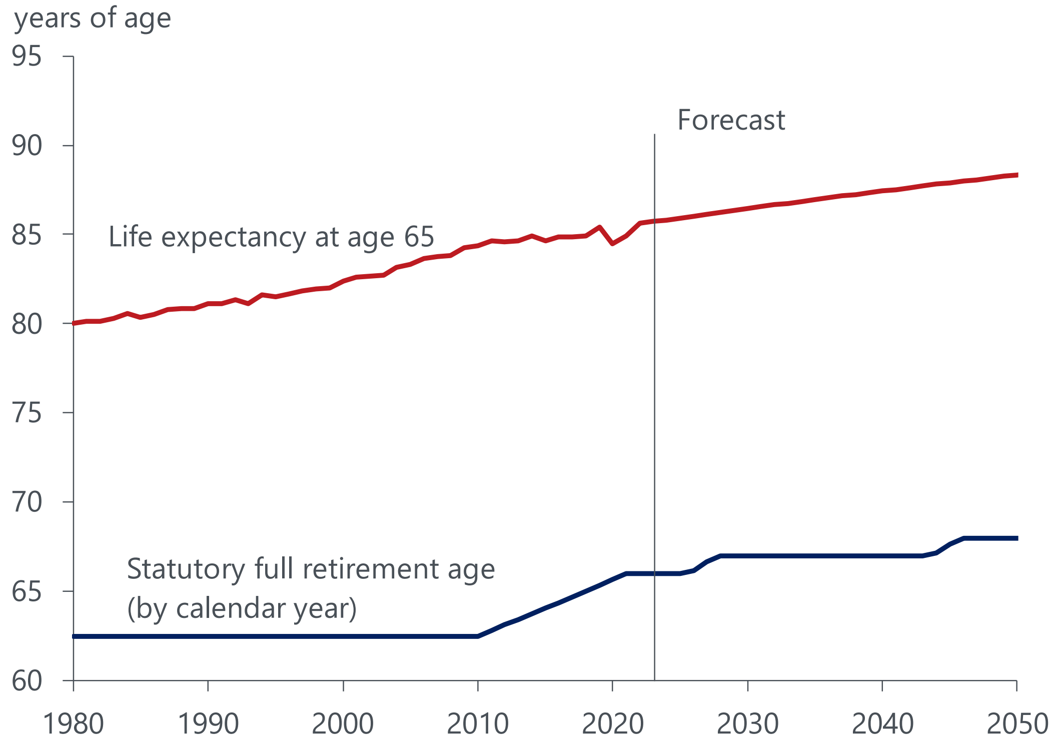


Source: Oxford Economics, Haver Analytics, UN, ONS

- Uses UN estimates spliced with ONS data.
- OE's forecast adopts the [ONS' low migration variant projection](#) as its baseline net migration assumption. This assumes net inflows of 253,000 a year over the medium-term.
- Impact on r*: Net migration slowly skews the population younger, a greater share of young households boosts loan demand relative to supply, raising r*.
- Unlike TFR, its impact is immediate, but slow given net migration is a flow variable.

Secular forces: Life expectancy and retirement age

UK: Life expectancy and retirement age

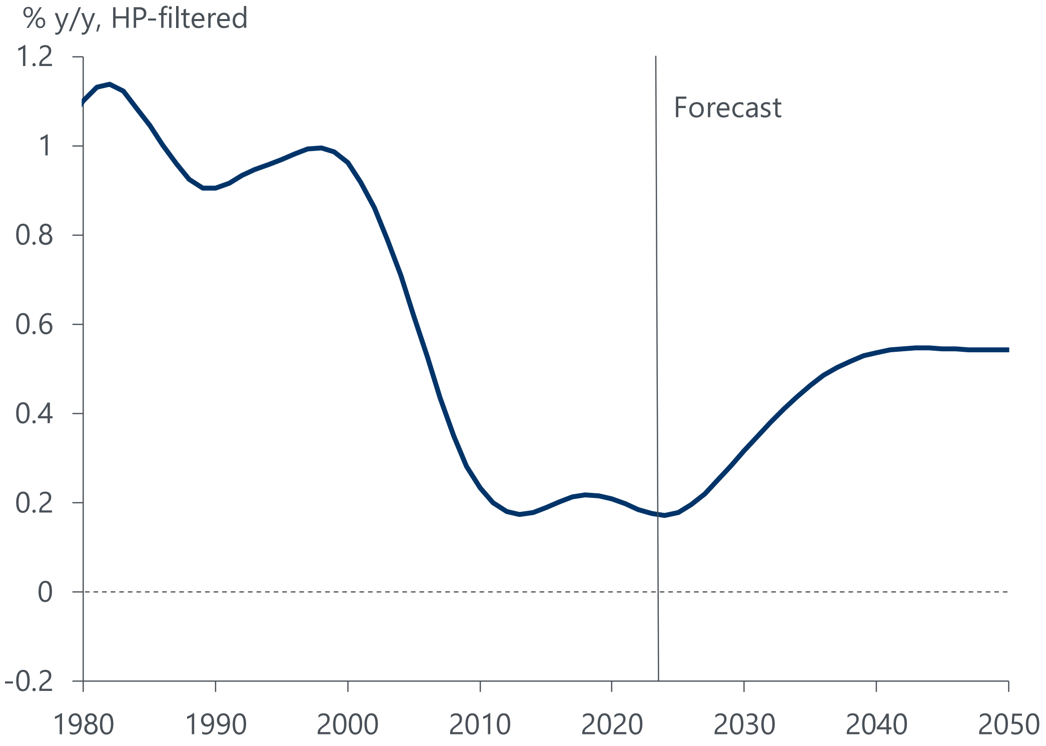


Source: Oxford Economics, ONS, UN WPP

- Statutory full retirement age is average of men and women. In model, adjusted by average effective time in retirement.
- Life expectancy is shown, but in the model, conditional survival rates by single year and age are used. Life expectancy is the sumproduct of those conditional survival rates.
- Impact on r* : The rising time span in retirement means more savings are accumulated during working life, shifting the capital supply curve to the right, depressing r*.

Secular forces: Total factor productivity

UK: Total factor productivity (TFP) trend

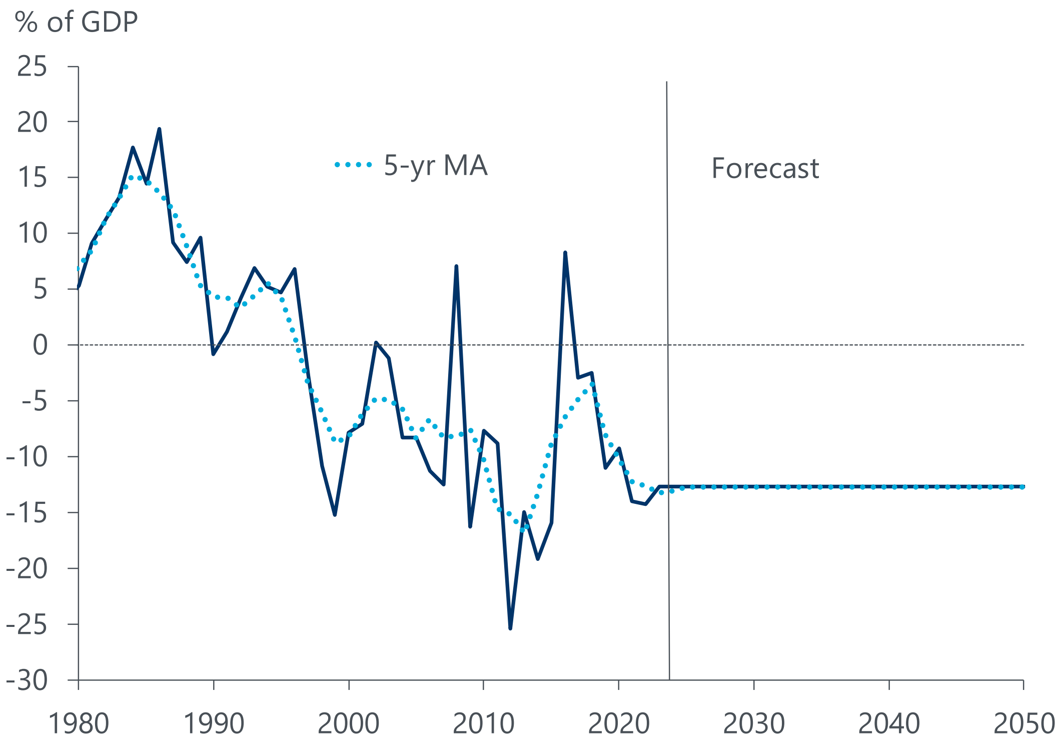


Source: Oxford Economics, Haver Analytics, ONS

- ONS estimates, spliced with OE forecast
- OE forecast available in OE databank
- HP-filtered (lambda=100)
- Impact on r*: lower TFP growth means lower
  - potential output growth
  - return to investment
  - wage growth
  - borrowing capability of young households
  - All of which depress the neutral rate

Secular forces: Net foreign assets

UK: Net foreign assets

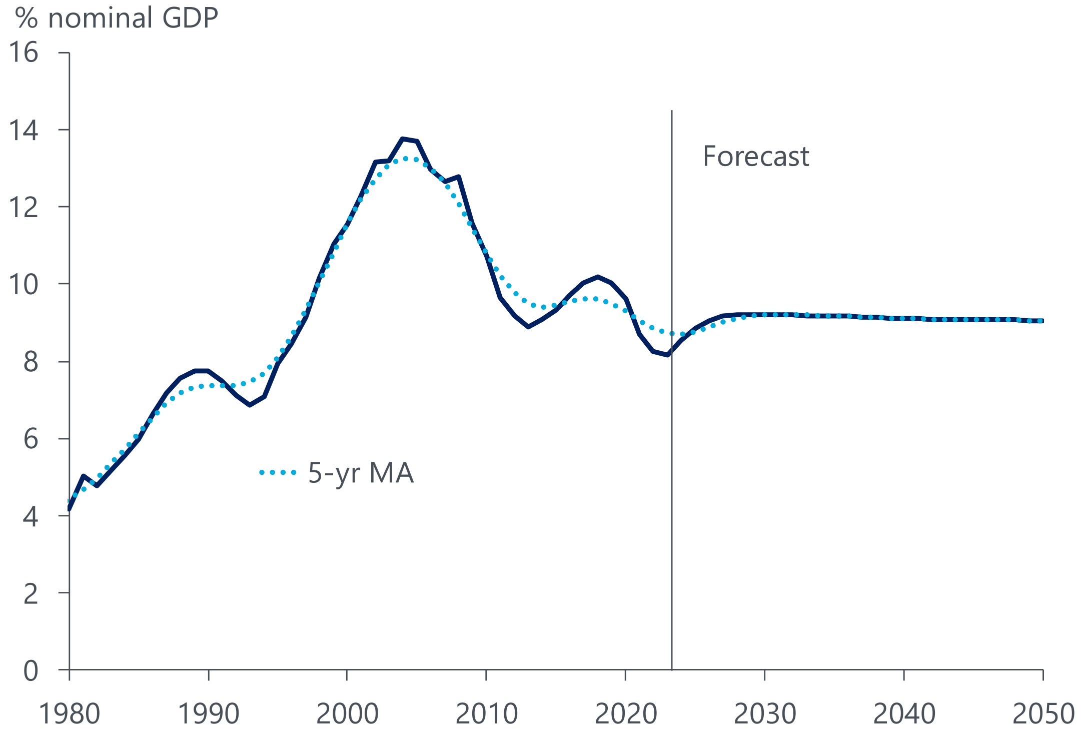


Source: Oxford Economics, Haver Analytics, ONS

- ONS estimates
- Impact on r*: capital inflows add to the domestic supply of savings, putting downward pressure on r*, and v.v. for outflows.

Secular forces: Consumer credit

UK: Consumer credit (ex student loans)

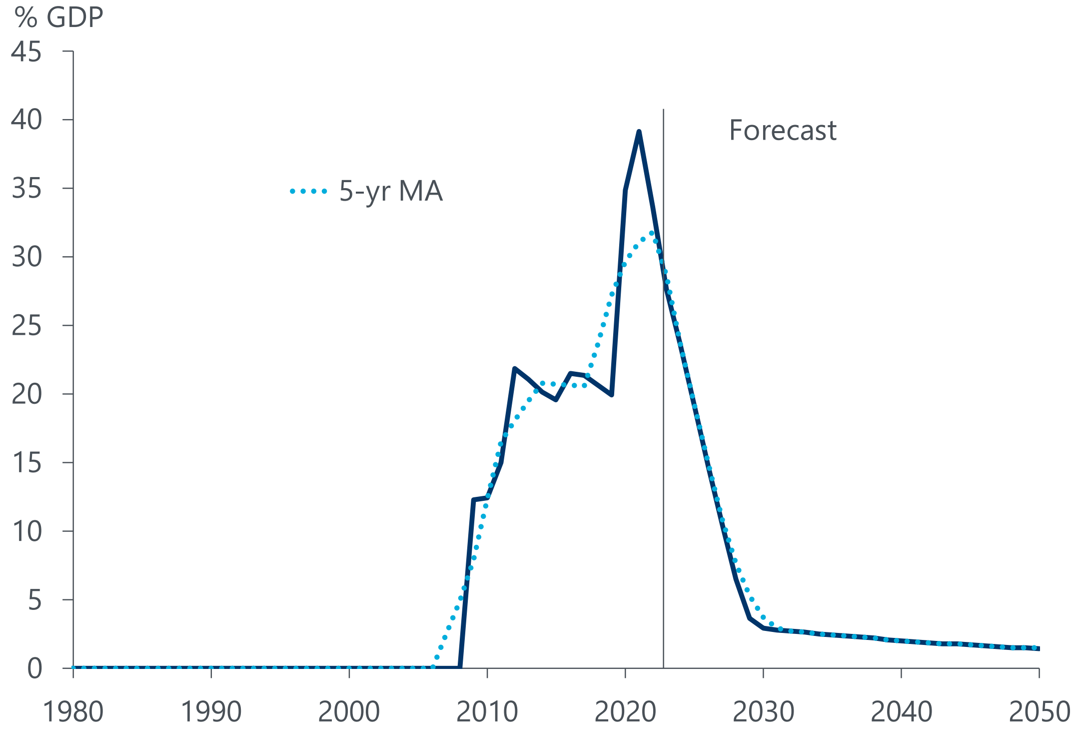


Source: Oxford Economics, Haver Analytics, BoE

- History from Bank of England
- Impact on r*:
  - As consumer credit rises, there's more demand for capital (in the form of loans).
  - This raises the neutral rate.
  - In model, this is implemented via credit constraints, representing financial deepening in the 1990s.

Secular forces: Stock of QE securities

UK: Stock of securities under QE programs

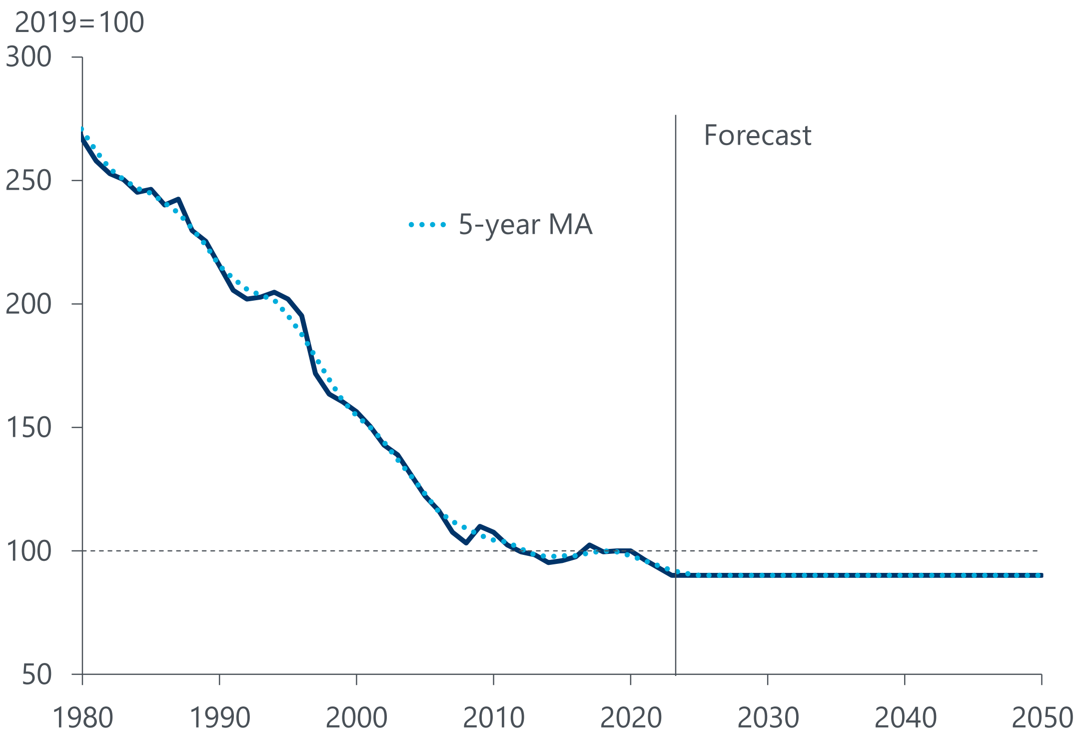


Source: Oxford Economics, Haver Analytics

- Securities on BoE's balance sheet
- OE forecast
- A new secular force because of magnitude out to 2050
  - For US and EZ, OE forecast remains at very elevated levels out to 2050.
- Impact on r*:
  - Reduces amount of outstanding government debt securities, thereby rasing r*.
  - Market segmentation/preferred habitat assumption. Cf. this Fed working paper.

Secular forces: Relative price of investment goods

UK: Relative price of equipment goods

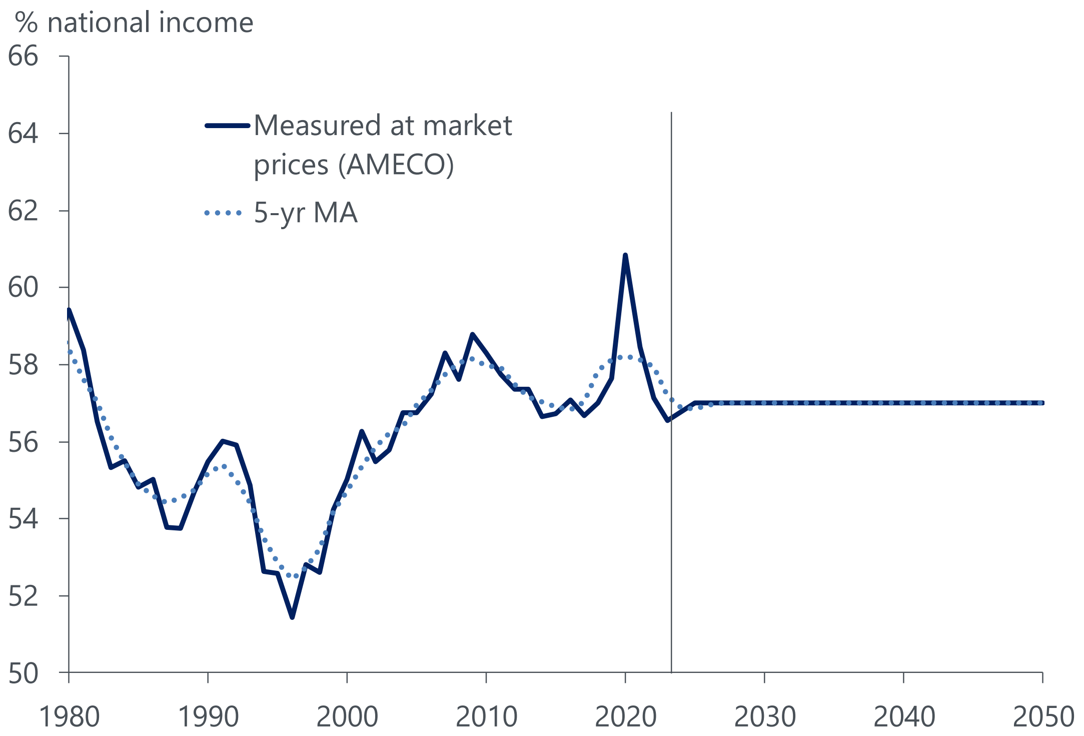


Source: Oxford Economics, Haver Analytics

- Computed analogous to Fernald (2014)
- Impact on r*:
  - As investment goods become relatively cheaper, less savings are needed to finance a given level of real investment in the capital stock.
  - Raises real capital supply, depressing the neutral rate.

Secular forces: Labour share of national income

UK: Labour share of income

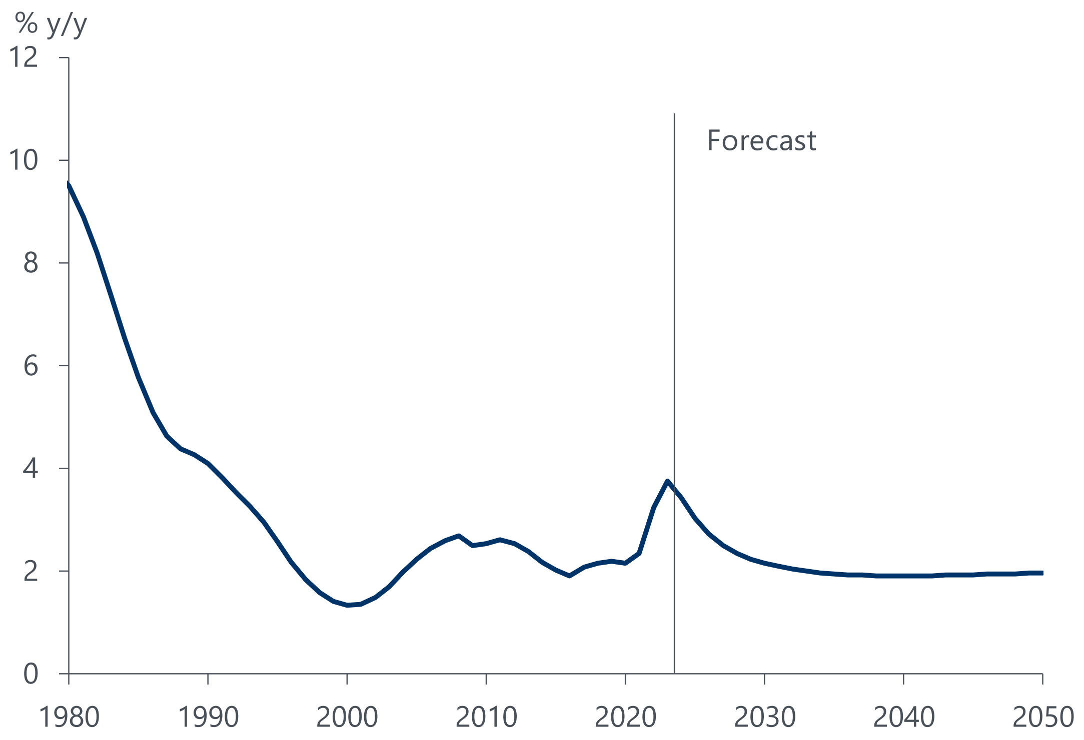


Source: Oxford Economics, Haver Analytics

- History from AMECO
- Impact on r*:
  - If labour share falls, more capital income accrues to capital owners, who have higher propensity to save
  - This raises capital supply and depresses the neutral rate
  - In model, the rise in the capital share of income results from an increase in profits, which arise due to monopolistic competition in the production of differentiated final goods

Inflation expectations corresponding to long-run neutral rate

UK: Inflation expectations for long-run neutral rate



Source: Oxford Economics, Haver Analytics, NIESR, Citi/YouGov

- Only needed for the nominal neutral rate, not for analysis of real rate
- Long-run trend of annual inflation expectations
- Spliced three series:
    - NIESR one-year ahead forecast
    - Citi/YouGov inflation expectations 1yr
    - OE core inflation forecast
- One-sided HP-filter to extract trend
    - Agents are backward looking
    - Sticky expectations
    - Avoid over-smoothing of two-sided filter
- Core CPI forecast available in OE Global Economic Databank

Parameter estimates and calibration

Estimated parameters		
Parameters	**Empirical targets**	**Parameter value**
Discount factor	nom. investm.-output ratio in 2019	0.96
Elasticity of intertemporal substitution	real rate of interest in 1996	0.7
Capital share in intermediate goods	real rate of interest in 2019	0.21
Bequest motive	bequest to income ratio in 2019	29.47
Borrowing limit in 2019	consumer-credit to GDP ratio 2019	0.23
Retailer elast. of substitution in 2019	labour share 2019	4.13
Borrowing limit in 1980	consumer-credit to GDP ratio 1980	0.15
Retailer elast. of substitution in 1980	labour share 1980	4.39
Key calibrated parameters		
Parameter	**Source**	**Parameter value**
Depreciation rate (ex structures)	Jorgenson (1996)	0.12
Production elasticity of substitution	Cesa-Bianchi et al. (2022)	0.7
Life-cycle productivity profile	Gourinchas and Parker (2002)	vector

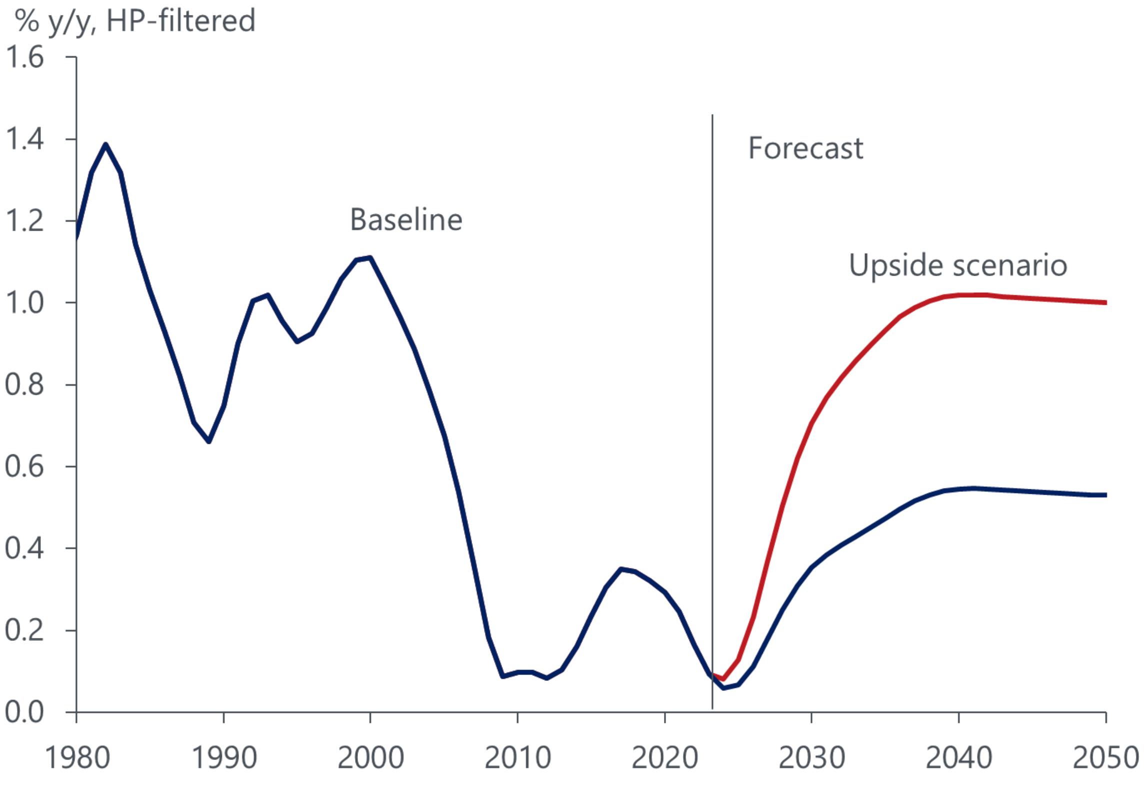
Source: Oxford Economics

- Estimated with method of simulated moments
  - Which is a restricted GMM
  - Model parameters estimated jointly to match all empirical targets
- Reference year in estimation: 2019, because policy rate roughly neutral (using methodology of Bofinger and Haas, 2023)
- Improved EIS estimation: calibrated to real rate in 1996 (Bofinger and Haas, 2023), within bounds by Ingemann & Facino (2017)
- For comparison with other estimates in literature, see our White Paper, p. 34.
- Full estimation takes ~3 hrs, approx 5000 lines of Matlab code

Scenarios

Scenario 1: High TFP growth

UK: TFP growth scenario

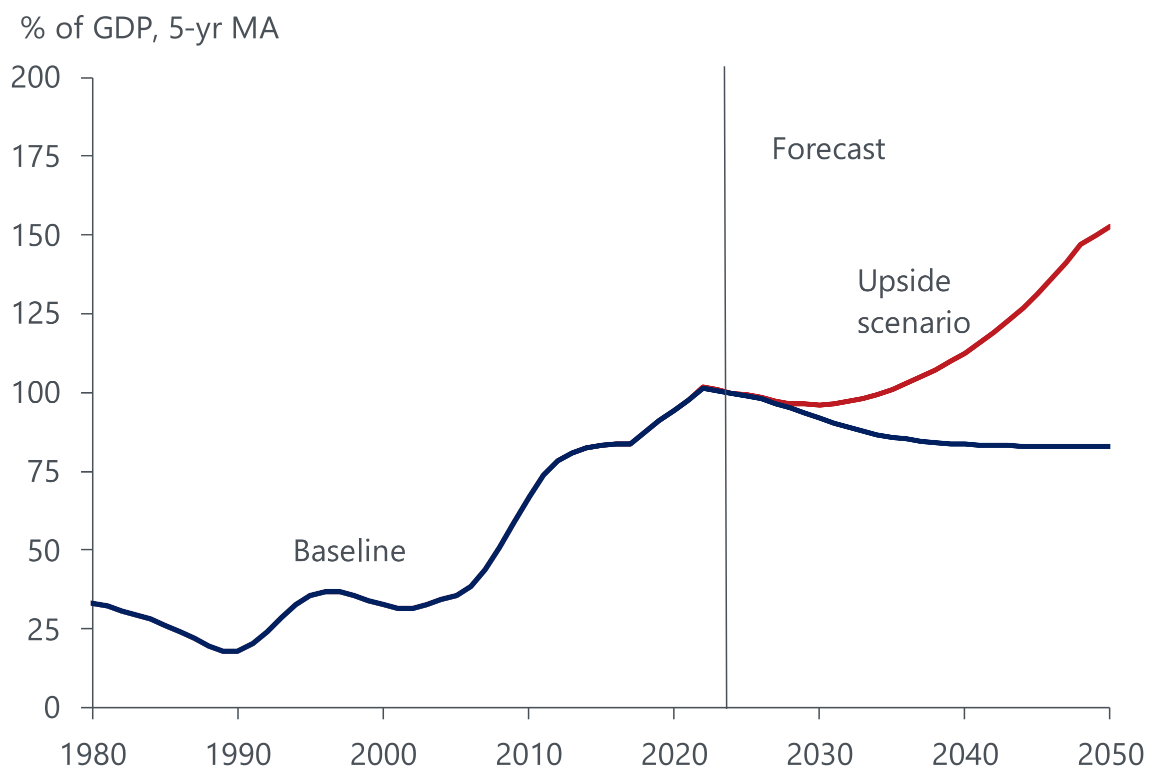


Source: Oxford Economics, ONS

- Higher productivity scenario assumes trend TFP growth returns to its 1980-2000 average rate of 1% y/y.
- Results: r* rises to -0.5% by 2050 vs -1.1% in baseline.
- Economic intuition:
  - Higher TFP growth ➔ higher output and investment growth ➔ upward pressure on r*.
  - Higher TFP growth ➔ stronger wage growth and borrowing capacity of younger households ➔ greater demand for loans relative to supply ➔ upward pressure on r*.

Scenario 2: Higher government debt

UK: Government debt scenario

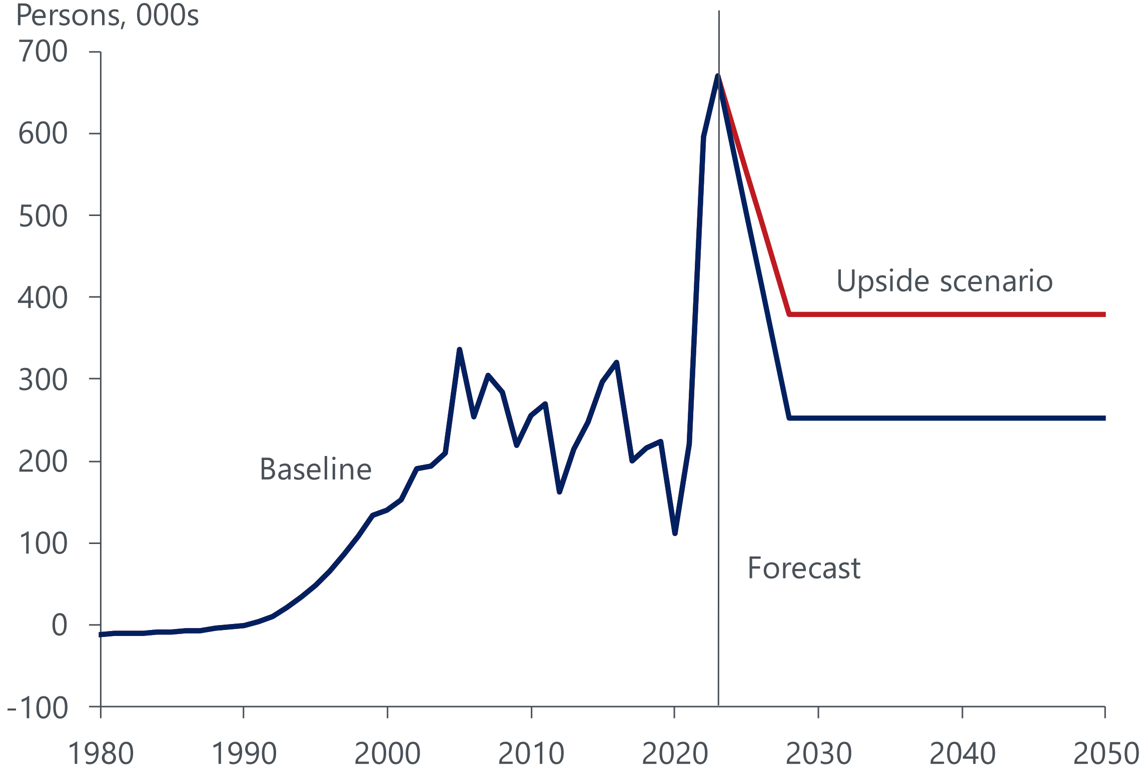


Source: Oxford Economics, OBR, Bank of England, Haver

- Higher government debt scenario uses the OBR's 2023 FSR departmental spending pressure scenario.
- Follows OBR's baseline, but assumes the departmental expenditure limit rises by 1.5% of GDP from 2029, resulting in a permenently lower primary balance.
- Outstanding government debt rises to around 160% of GDP by 2050, instead of the 84% in our baseline.
- Results: r* rises to 1.6% by 2050 vs -1.1% in baseline.
- Larger deficits lead to upward pressure on r*:
  - Governments have a greater demand for capital to finance deficits.

Scenario 3: Higher net inward migration

UK: Net migration scenario

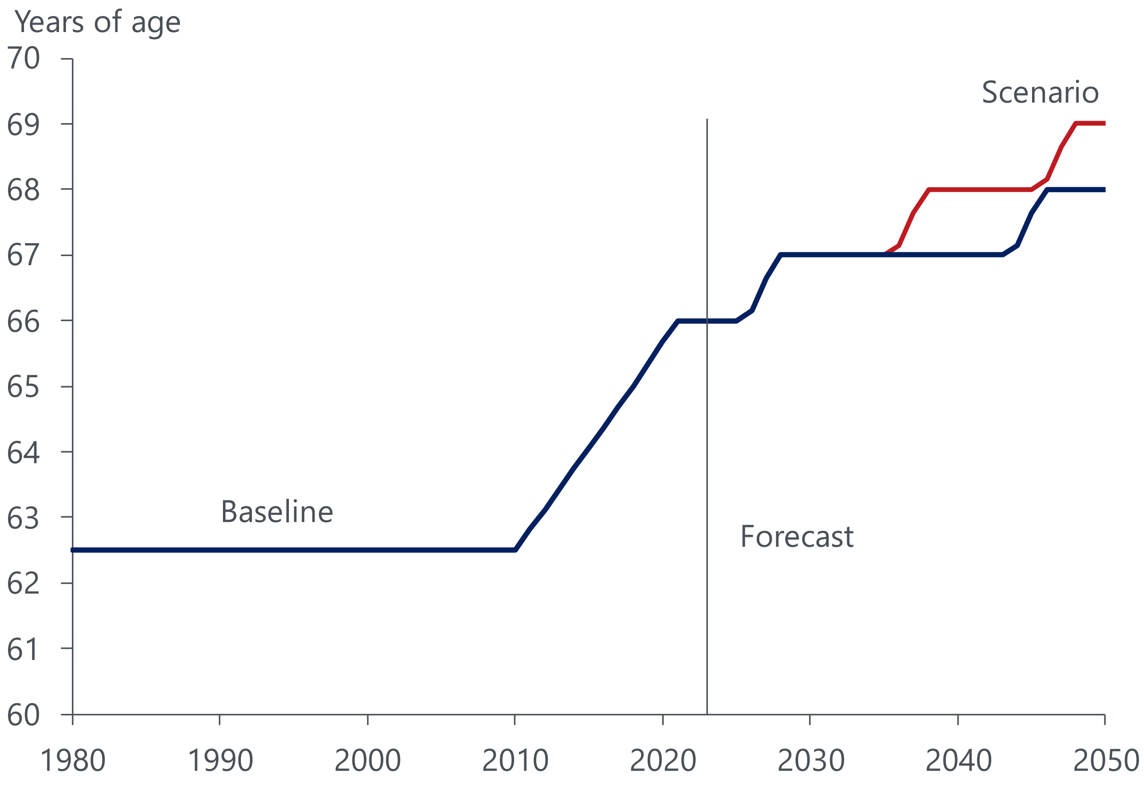


Source: Oxford Economics, Haver Analytics, UN, ONS

- Higher net migration scenario follows [ONS' higher net migration variant] of 379k people a year instead of the ONS' low variant (253k) used in our baseline.
- Results: r* rises to -0.6% by 2050 vs -1.1% in baseline.
- Economic intuition: Higher net inward migration ➔ labour force slowly skews younger ➔ greater share of young households boosts loan demand relative to supply ➔ upward pressure on r*.
- Impact on r* unfolds over time because net migration is an annual flow variable. Therefore, its initial influence is small, but gradually builds over time.

Scenario 4: Higher statutory retirement age

UK: Scenario higher statutory retirement age



Source: Oxford Economics, Haver Analytics

- Our baseline follows current government policy of implementing two single-year increases to the statutory state pension age (in 2026-28 and 2046-48).
- Higher retirement age scenario incorporates an additional single-year increases to the statutory state pension age in 2036-38.
- Results: r* rises to -0.6% by 2050 vs -1.1% in baseline.
- Economic intuition: Higher retirement age ➔ shorter period spent in retirement ➔ households accumulate relatively less savings in preparation ➔ lower supply of capital ➔ upward pressure on r*

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EUROPE

Oxford (Headquarters)
Tel: +44 (0)1865 268 900

London
Tel: +44 (0)20 3910 8000

Belfast
Tel: + 44 2892 635400

Milan
Tel: +39 02 8295 2521

Frankfurt
Tel: +49 69 96 758 658

Paris
Tel: +33 (0)1 78 91 50 52

Stockholm
Tel: +46 (0) 8 446 887 65

AFRICA AND MIDDLE EAST

Cape Town
Tel: +27(0)21 863-6200

Dubai
Tel: +971 56 396 7998

AMERICAS

New York
Tel: +1 (646) 786 1879

Philadelphia
Tel: +1 (646) 786 1879

Mexico City
Tel: +52 155 5419-4173

Boston
Tel: +1 (617) 780 2265

Chicago
Tel: +1 (847) 993-3140

Los Angeles
Tel: +1 (424) 303 3449

Toronto
Tel: +1 (905) 361 6573

ASIA PACIFIC

Singapore
Tel: +65 6850 0110

Sydney
Tel: +61 (0)2 8458 4200

Hong Kong
Tel: +852 3974 8842

Tokyo
Tel: +81-(0)3-4588-2798